

Briefcase Requirements Specification

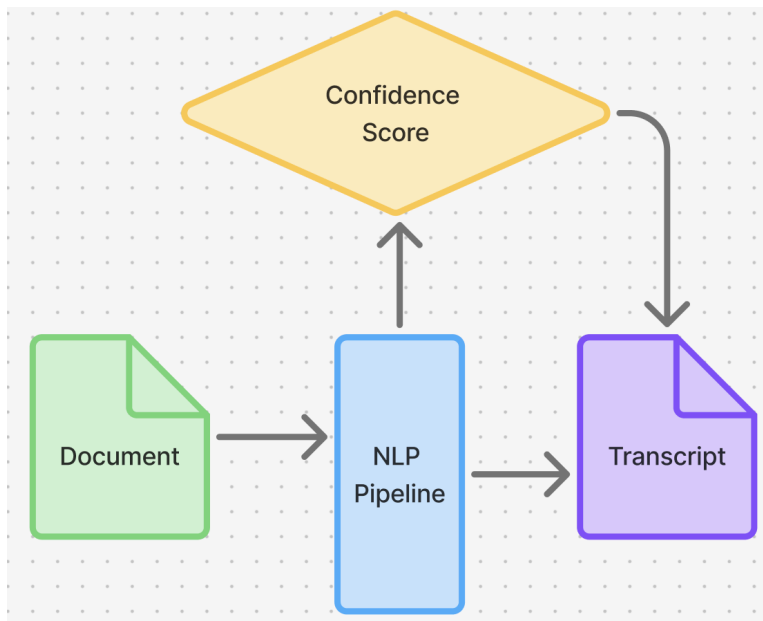
Fall 2026 Senior Project

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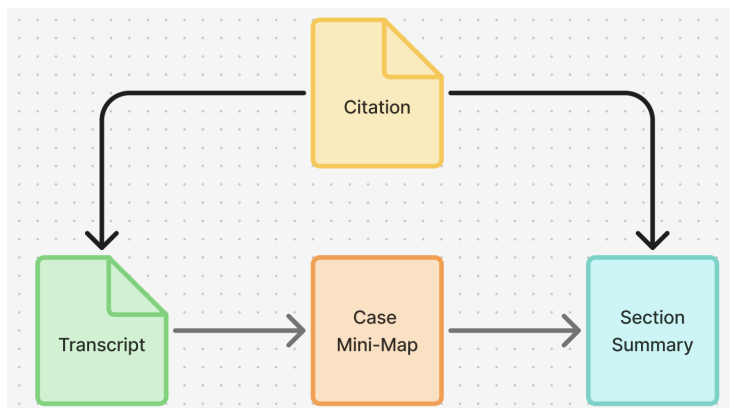
1. Introduction

The goal of this project is to increase user transparency. The Briefcase software includes: text extraction which will be improved with an associated confidence score, case minimap summaries which will be improved with more accurate summaries and attached citations and document chat which will be improved with response citations. This document includes the requirements specification for the Briefcase transparency project.

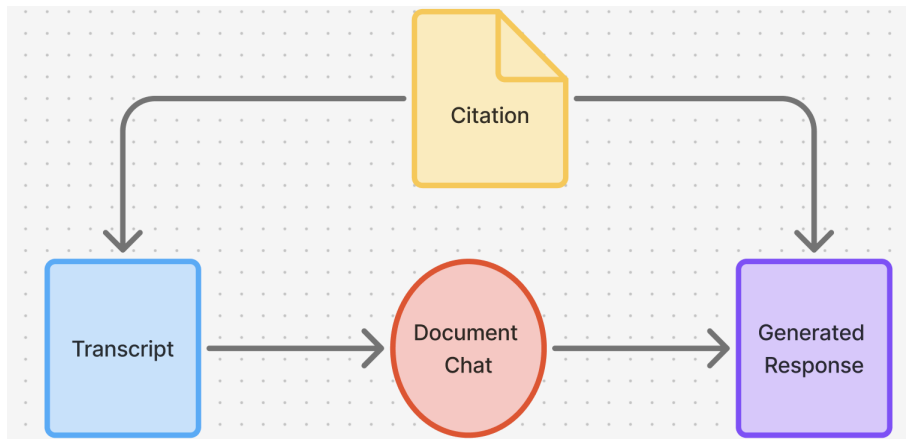
2.1 Text Extraction



2.2 Case Minimap System



2.3 Document Chat Subsystem



2. Subsystem Overview

Briefcase is composed of the following subsystems (that we will be working on):

2.1. Text Extraction Subsystem

Optical Character Recognition – uses Tesseract OCR to recognize characters on scanned PDFs

Clustering System – separates out yeetable garbage from actual text

2.2. Case Minimap Subsystem

Section Header – topical titles/headers describing different sections/examinations

Summary – AI-generated summary of examination

Section Citation – text pointing to the transcript line number where the summary sourced a claim from

2.3. Document Chat Subsystem

Input Panel – user submitted text asking questions

Response Field – generated answers drawn from transcript

Response Citation – additional text added onto the generated answers showing where answers come from

3. Text Extraction Subsystem

The following requirements apply to the text extraction subsystem for Briefcase:

3.1. As a lawyer, I want to view a confidence score for extracted text so that I can better understand how reliable the extracted text is.

- 3.1.1. Acceptance criteria: When the user clicks on a sentence of extracted text, the system displays the confidence score associated with that sentence.
- 3.1.2. Acceptance criteria: The confidence score is displayed as a percentage between 0% and 100%.
- 3.1.3. Acceptance criteria: A score of 100% represents the highest confidence that the extracted text is correct.
- 3.1.4. Acceptance criteria: The confidence score is displayed next to or otherwise clearly associated with the selected sentence.

- 3.1.5. Acceptance criteria: The confidence score appears within 3 seconds of the user selecting the sentence.
- 3.1.6. Acceptance criteria: The system assigns a confidence score to at least 80% of extracted text.
- 3.1.7. Acceptance criteria: If a confidence score is unavailable for the selected text, the system indicates that no score is available rather than displaying an inaccurate or placeholder score.
- 3.2. As a lawyer, I want the confidence score to disappear when I click outside of the selected text so that I can review other portions of the document without the score remaining visible.
 - 3.2.1. Acceptance criteria: When the confidence score is displayed and the user clicks outside of the selected sentence, the confidence score disappears.
 - 3.2.2. Acceptance criteria: Clicking on a different sentence updates the displayed confidence score to correspond to the newly selected sentence.
 - 3.2.3. Acceptance criteria: The confidence score does not remain visible after the user has clicked outside the selected sentence.
- 3.3. As a lawyer, I want low-confidence extracted text to be highlighted so that I can quickly identify portions of the document that may contain OCR errors.
 - 3.3.1. Acceptance criteria: Text with a confidence score below 75% is automatically highlighted.
 - 3.3.2. Acceptance criteria: Text with a confidence score of 75% or higher is not highlighted as low-confidence text.
 - 3.3.3. Acceptance criteria: Highlighting is applied automatically when the document is displayed or when text extraction is completed.
 - 3.3.4. Acceptance criteria: Highlighted text remains associated with the corresponding extracted text and does not change the document's extracted content.
- 3.4. As a lawyer, I want the highlighting to indicate the severity of potential OCR errors so that I can quickly identify areas of the document that require closer review.
 - 3.4.1. Acceptance criteria: Text with a confidence score from 50% through 74.99% is highlighted in orange.
 - 3.4.2. Acceptance criteria: Text with a confidence score below 50% is highlighted in red.
 - 3.4.3. Acceptance criteria: Text with a confidence score of 75% or higher is not highlighted.
 - 3.4.4. Acceptance criteria: The highlight color is determined automatically based on the confidence score.
 - 3.4.5. Acceptance criteria: The same confidence-score ranges are applied consistently throughout the document.
- 3.5. As a lawyer, I want to turn accuracy highlighting on and off so that I can choose whether potential OCR errors are displayed while reviewing a document.

- 3.5.1. Acceptance criteria: The interface provides a clearly labeled control for turning accuracy highlighting on and off.
- 3.5.2. Acceptance criteria: Accuracy highlighting is enabled when the user turns the control on.
- 3.5.3. Acceptance criteria: Accuracy highlighting is removed when the user turns the control off.
- 3.5.4. Acceptance criteria: Turning highlighting off does **not** remove or modify the extracted text.
- 3.5.5. Acceptance criteria: Turning highlighting back on restores the appropriate highlights based on the confidence scores.
- 3.5.6. Acceptance criteria: The confidence score remains available when a user selects extracted text, regardless of whether accuracy highlighting is enabled or disabled.

4. Document Chat Subsystem

The following requirements apply to the DocChat subsystem for Briefcase:

- 4.1. As a lawyer I want the responses given by the document chat to include transcript documentation citations so that I can verify the accuracy of the given response.
 - 4.1.1. Acceptance criteria: The chat must include citations for more than 90% of responses.
 - 4.1.2. Acceptance criteria: The citation must appear alongside the given response in the document chat.
- 4.2. As a lawyer I want the citation to include the possible multiple page numbers which were used to generate the response so that I can manually verify the response.
 - 4.2.1. Acceptance criteria: The citations should include page number for more than 90% of generated responses.
 - 4.2.2. Acceptance criteria: The citations should display the page number(s) which was used to generate a response.
- 4.3. As a lawyer I want the page numbers in the citations to be accurate directly leading to the page(s) which were used to generate the response so that I can know that the generated citation corresponds to the page(s) in the document transcript which was used to generate the response.
 - 4.3.1. Acceptance criteria: The citations should have an accuracy of more than 90%
- 4.4. As a lawyer I want the citation to be a link which will take me to the transcript document pages inside of the workspace so that I do not have to scroll through potentially hundreds of pages to look for the pages generated in the citation.
 - 4.4.1. Acceptance criteria: The citations should be clickable and take the user to the transcript page.

- 4.4.2. Acceptance criteria: The citation link should lead to the transcript page more than 90% of the time.

5. Case Minimap Subsystem

The following requirements apply to the case minimap subsystem for Briefcase:

- 5.1. As a lawyer, I want the minimap to break down the structure of the trial into descriptive headers so I can find the information I'm looking for without having to scroll through thousands of pages by hand.
 - 5.1.1. Acceptance criteria: The minimap shall organize transcript content into visually distinct, labeled sections.
 - 5.1.2. Acceptance criteria: Each identified examination shall display a header containing, at minimum, the witness name and examination type when that information is available in the transcript.
 - 5.1.3. Acceptance criteria: Examination types shall distinguish between direct examination, cross-examination, redirect examination, and recross examination when explicitly identified in the transcript.
 - 5.1.4. Acceptance criteria: Sections shall appear in the same chronological order in which they occur in the source transcript.
 - 5.1.5. Acceptance criteria: Each section header shall be visually distinguishable in a different typeset from the summary content associated with that section.
- 5.2. As a lawyer, I want the minimap to have full coverage of the trial transcript so that I have the complete story of the case just from reading the overview.
 - 5.2.1. Acceptance criteria: The minimap shall generate an entry for every examination identified in the submitted transcript.
 - 5.2.2. Acceptance criteria: Each minimap entry shall correspond to a specific section of the source transcript.
 - 5.2.3. Acceptance criteria: The order of minimap entries shall match the order of the corresponding sections in the source transcript.
 - 5.2.4. Acceptance criteria: If a transcript section cannot be processed, the minimap shall display an indication that the section could not be summarized rather than silently omitting it.
- 5.3. As a lawyer, I want the minimap to concisely summarize the contents of each examination so that I can get a condensed overview of what happened.
 - 5.3.1. Acceptance criteria: A generated summary shall be displayed for each successfully processed examination.
 - 5.3.2. Acceptance criteria: Each examination summary shall contain no more than 200 words.
 - 5.3.3. Acceptance criteria: The summary shall identify the primary subjects or events discussed during the examination.
 - 5.3.4. Acceptance criteria: For cases within the system's supported document-size limit, the generated minimap shall be available within 10 seconds after transcript processing is completed.

- 5.4. As a lawyer, I want each minimap summary to contain citations pointing to lines in the trial transcript so that it is more transparent about where claims are being sourced from.
 - 5.4.1. Acceptance criteria: Each factual claim in an examination summary shall contain at least one citation to supporting transcript content.
 - 5.4.2. Acceptance criteria: Each citation shall identify the page and/or line range containing the supporting source text, based on the citation information available in the trial transcript.
 - 5.4.3. Acceptance criteria: Citation references shall correspond to the transcript associated with the currently displayed case.
 - 5.4.4. Acceptance criteria: A citation shall not reference transcript content outside the submitted case.
 - 5.4.5. Acceptance criteria: When the system cannot identify supporting transcript content for a generated statement, that statement shall not be presented as a supported factual claim.
- 5.5. As a lawyer, I want to be able to click on the citation text within the summary to redirect me to its location in the original document so that navigation is easier.
 - 5.5.1. Acceptance criteria: Each transcript citation displayed in the minimap shall be selectable.
 - 5.5.2. Acceptance criteria: Selecting a citation shall navigate the transcript viewer to the cited page or line range.
 - 5.5.3. Acceptance criteria: The cited text or cited line range shall be visually highlighted after navigation.
 - 5.5.4. Acceptance criteria: Citation navigation shall not close or reset the minimap.
- 5.6. As a lawyer, I want to see a confidence score of the accuracy of each summary next to each minimap section so that I can better visualize the reliability of the information being presented.
 - 5.6.1. Acceptance criteria: Each generated examination summary shall display a confidence score adjacent to the summary.
 - 5.6.2. Acceptance criteria: Confidence shall be displayed using a consistent numerical scale, such as 0–100%.
 - 5.6.3. Acceptance criteria: The displayed confidence score shall correspond to the specific examination summary with which it is displayed.
 - 5.6.4. Acceptance criteria: Confidence scores shall be constrained to values between 0% and 100%.
 - 5.6.5. Acceptance criteria: The confidence score shall be recalculated whenever the corresponding summary is regenerated.
- 5.7. As a lawyer, I want to see a clear flag/icon by sections where confidence scores are below a threshold so there is a clear disclaimer when the confidence of a summary is low and should be taken with a grain of salt.

- 5.7.1. Acceptance criteria: The system shall define a configurable confidence threshold for identifying low-confidence summaries.
- 5.7.2. Acceptance criteria: Any summary with a confidence score below the configured threshold shall display a warning icon adjacent to its confidence score.
- 5.7.3. Acceptance criteria: Summaries with confidence scores equal to or above the threshold shall not display the low-confidence warning.
- 5.7.4. Acceptance criteria: The warning shall include accessible text indicating that the summary has low confidence and should be verified against the source transcript.
- 5.7.5. Acceptance criteria: The warning shall not prevent the user from viewing the summary, citations, or original transcript.
- 5.7.6. Acceptance criteria: The warning shall not rely solely on color to communicate low confidence.

6. Briefcase Environment Requirements

The following software and resources needed to develop and run Briefcase's extraction, case minimap, and document chat components.

6.1. Development Environment Requirements

The following are the Development Environment Requirements for this portion of Briefcase:

Software updates will use Briefcase's editing technology stack:

Category	Requirement
Frontend	React, Typescript
Backend	Python, Flask
Database and related services	Firebase, Elasticsearch, AWS RDS, PostgreSQL
Text Extraction	Tesseract OCR, Scikit-learn
Structure and Relationships	spaCy
Higher-Level Answers	PyTorch, Scikit-lern, Hugging Face
Infrastructure	Docker

Tesseract OCR identifies and recognizes text from scanned documents, while Scikit-learn supports the clustering system used to identify which parts of a page to process. spaCy supports part-of-speech tagging, namely entity recognition, and uses rule based matching to organize the extracted text.

6.2. Execution Environment Requirements

The execution environment requirements must be managed via the dependency files located in the project.

6.3. Security and Privacy Environment Requirements

Briefcase handles sensitive case information, including potentially active cases. The project is required to be HIPAA compliant as requested by Loyola's Project for the Innocent (LPI) clinic. To meet this requirement, no case information can be submitted to or accessed by public AI models that may use the data for training. These restrictions apply to all tools and services that are used to process case documents.